# Pavan Kumar S

**Executive Business Name: Pavan Kumar Sadashiv | Founder & MD, HRL International Private Limited**
**AI Systems Chief Architect | GPU Systems & High-Performance Runtime Engineer**
Mangaluru, Karnataka, India | Email: hrlinternationalprivatelimited@gmail.com | GitHub: github.com/hrlpavan
Platform: hrlpavan.github.io/hrl-international-website- | Charter: github.com/hrlpavan/engineering-compensation-blueprint

## PROFESSIONAL SUMMARY

High-velocity AI Systems Chief Architect and GPU Systems Engineer with specialized expertise in low-level CUDA/C++20 high-performance inference runtimes, Paged KV-Cache virtualization, CUDA Graph optimization, hardware-aware FP8/INT4-AWQ quantization, and discrete combinatorial optimization (NP-hard CSP Backtracking). Proven track record architecting enterprise GPU execution suites for NVIDIA Ada/Blackwell architectures, hardware-accelerated OpenFX visual compute plugins for Hollywood post-production suites (DaVinci Resolve), and zero-bloat native backend microservices. Founder and Managing Director of HRL International Private Limited. Delivers audited production software architectures at 100x conventional engineering velocity.

## CORE TECHNICAL COMPETENCIES

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| **GPU & Low-Level Systems:** | C++20/17, NVIDIA CUDA 12.6, Apple Metal (MSL), CUDA Graphs, Paged Attention & KV-Cache, OpenFX API, GPU Memory Optimization |
| **AI/ML & Inference Optimization:** | PyTorch, ONNX Runtime, FP8 (E4M3/E5M2) & INT4-AWQ Quantization, Speculative Decoding, Depth Anything AI, ACEScc Color Science |
| **Algorithms & Discrete Optimization:** | Combinatorial Optimization, NP-Hard Constraint Satisfaction (CSP) Backtracking, Dynamic K-Means (k=3), Bayesian Classification |
| **Backend & Systems Architecture:** | Native Node.js (Zero-Framework Architecture), SQLite 3 (ACID Transactions), REST APIs, Google Antigravity SDK, Cloud Composer, BigQuery |
| **DevOps & Build Systems:** | CMake, Git, GitHub Actions (CI/CD), macOS DMG Application Bundling, Linux Kernel/Driver Environments |

## FLAGSHIP ENGINEERING & GPU PROJECTS

### 1. NVIDIA RTX-LocalAI Enterprise Execution Suite

*Technologies: C++20, CUDA 12.6, Paged KV-Cache, CUDA Graphs, Speculative Decoding, INT4-AWQ, FP8, CMake* | hrlpavan.github.io/nvedia-project-by-hrl/

- Architected a C++20/CUDA 12.6 on-device AI inference engine and memory management subsystem for NVIDIA Ada Lovelace / Blackwell & DGX systems.
- **Paged KV-Cache Virtualization**: Engineered a block manager (16 tokens/block) achieving **0.17 µs/request latency** with **0.00% memory fragmentation** on 8GB–24GB VRAM.
- **CUDA Graphs & Continuous Batching**: Pre-captured execution graphs dropping CPU launch latency to **< 5 µs** with iteration-level continuous batching (512 tokens/chunk).
- **Speculative Policy Engine & FP8/INT4 Kernels**: Implemented draft-target verification loops for **up to 2.2x speedup** (88.4% acceptance) and custom dequant GEMM/GEMV CUDA kernels.

### 2. AI Cinematic Haze — OpenFX & Fusion GPU Plugin for DaVinci Resolve

*Technologies: C++17/20, NVIDIA CUDA, Apple Metal (MSL), OpenFX API, Depth Anything AI, ONNX Runtime, ACEScc* | github.com/hrlpavan/hrl-international-website-

- Architected a high-throughput OpenFX visual compute plugin with native CUDA and Apple Metal backends for DaVinci Resolve 21.
- Integrated monocular neural depth estimation (Depth Anything via ONNX Runtime) to compute real-time volumetric optical atmospheric scattering.
- Preserved uncompressed 12-bit DPX sensor dynamic range through strict Hollywood ACEScc color science management.

### 3. InvigiMatrix — College Examination & Invigilation Management Portal

*Technologies: JavaScript (ES6+), Native Node.js (Zero-Framework), SQLite 3, CSP Backtracking, K-Means Clustering* | github.com/hrlpavan/exam-invigilation-dsa-app

- Implemented an NP-hard Constraint Satisfaction Problem (CSP) Backtracking solver eliminating scheduling collisions across faculty and venues.
- Built an AI/ML Smart Batch Planner with dynamic K-Means (k=3) & Bayesian classifier for anti-cheating student seat dispersion.
- Engineered a zero-framework native Node.js REST server backed by SQLite with strict ACID-compliant transactions.

## PROFESSIONAL EXPERIENCE & EDUCATION

**Founder & Managing Director — HRL International Private Limited (Pavan Kumar Sadashiv) (2026 – Present)**

- Founded enterprise deep-tech software venture; authored Startup India Seed Fund & Karnataka ELEVATE grant dossiers.
- Author of *Rule Breaking (Volume 4.0): The 100x AI Chief Architect Manifesto* (Master Treatise).

**Bachelor of Engineering (B.E.) in Computer Science & Engineering (AI & ML)**

Sahyadri College of Engineering & Management (SCEM), Mangaluru, Karnataka, India

- **Certifications:** JPMorgan Chase (Distributed Streaming/Kafka) • Deloitte Australia (Cybersecurity) • Blackmagic Design (Color Science)